

ISRO 2025: Topic-Wise Practice Questions

TOPIC 1: DATA STRUCTURES & ALGORITHMS

Based on 2023 ISRO Questions: 1, 3, 4, 5, 6, 24, 25, 26, 27, 28, 29, 46

****Coverage Analysis****: This topic had 12 questions (15%) in 2023 paper, making it the highest weightage area.

****Key Concepts Tested****: Graph algorithms (MST, shortest paths), tree traversals, expression evaluation, complexity analysis, sorting algorithms, array operations.

GRAPH ALGORITHMS (Questions 1-15)

****1. Find the minimum spanning tree cost for the graph with vertices {A,B,C,D} and edges: AB=3, AC=5, AD=7, BC=2, BD=4, CD=6.****

- (a) 9
- (b) 11
- (c) 13
- (d) 15

****2. Which algorithm would you use to find shortest path from one vertex to all others in a graph with negative edge weights:****

- (a) Dijkstra's algorithm
- (b) Bellman-Ford algorithm
- (c) Floyd-Warshall algorithm
- (d) Prim's algorithm

****3. In Kruskal's algorithm, edges are sorted in:****

- (a) Decreasing order of weights
- (b) Increasing order of weights
- (c) Random order
- (d) Order doesn't matter

****4. The time complexity of Dijkstra's algorithm using adjacency matrix is:****

- (a) $O(V^2)$
- (b) $O(V + E)$
- (c) $O(E \log V)$
- (d) $O(V \log V)$

****5. Floyd-Warshall algorithm is used to find:****

- (a) Minimum spanning tree
- (b) Single source shortest paths
- (c) All pairs shortest paths
- (d) Maximum flow

****6. For a connected graph with V vertices, minimum spanning tree will have:****

- (a) V edges
- (b) $V-1$ edges
- (c) $V+1$ edges
- (d) $2V$ edges

****7. Which data structure is used in Prim's algorithm for efficient implementation:****

- (a) Stack
- (b) Queue
- (c) Priority Queue
- (d) Array

****8. In a graph with V vertices and E edges, what is the maximum number of edges:****

- (a) $V(V-1)$
- (b) $V(V-1)/2$
- (c) V^2
- (d) $2V$

****9. Breadth-First Search uses which data structure:****

- (a) Stack
- (b) Queue
- (c) Priority Queue
- (d) Tree

****10. Depth-First Search time complexity is:****

- (a) $O(V + E)$
- (b) $O(V^2)$
- (c) $O(E \log V)$
- (d) $O(V \log V)$

****11. For detecting cycle in undirected graph, which approach is most efficient:****

- (a) DFS
- (b) BFS
- (c) Dijkstra's
- (d) Floyd-Warshall

****12. Topological sorting is possible only in:****

- (a) Undirected graphs
- (b) Directed acyclic graphs
- (c) Complete graphs
- (d) Bipartite graphs

****13. Which algorithm can handle negative edge weights:****

- (a) Dijkstra's algorithm
- (b) Prim's algorithm
- (c) Bellman-Ford algorithm
- (d) Kruskal's algorithm

****14. Time complexity of Floyd-Warshall algorithm is:****

- (a) $O(V^2)$
- (b) $O(V^3)$
- (c) $O(V^2 \log V)$
- (d) $O(V + E)$

****15. To find strongly connected components, which algorithm is used:****

- (a) Kruskal's
- (b) Kosaraju's
- (c) Dijkstra's
- (d) Prim's

**TREE DATA STRUCTURES (Questions 16-25)**

****16. Given preorder traversal (50, 30, 20, 40, 70, 60, 80), find the postorder traversal of this BST:****

- (a) (20, 40, 30, 60, 80, 70, 50)
- (b) (80, 70, 60, 50, 40, 30, 20)
- (c) (20, 30, 40, 50, 60, 70, 80)
- (d) (20, 40, 30, 80, 60, 70, 50)

****17. In a binary tree with n nodes, maximum height is:****

- (a) n
- (b) n-1
- (c) $\log n$
- (d) $n/2$

****18. Time complexity to find diameter of binary tree with n nodes:****

- (a) $O(n)$
- (b) $O(n^2)$
- (c) $O(\log n)$
- (d) $O(n \log n)$

****19. In a complete binary tree with n nodes, number of leaf nodes is:****

- (a) $\lceil n/2 \rceil$
- (b) $\lfloor n/2 \rfloor$
- (c) n-1
- (d) $\log n$

****20. For a BST with preorder (25, 15, 10, 22, 35, 30, 40), the inorder predecessor of 25 is:****

- (a) 15
- (b) 22
- (c) 10
- (d) 30

****21. Maximum number of nodes at level k in a binary tree is:****

- (a) 2^k
- (b) $2^{(k-1)}$
- (c) k

(d) $2k$

****22. In AVL tree, balance factor of any node is:****

- (a) $\{-1, 0, 1\}$
- (b) $\{-2, -1, 0, 1, 2\}$
- (c) Any integer
- (d) Only positive integers

****23. Time complexity of searching in a balanced BST:****

- (a) $O(n)$
- (b) $O(\log n)$
- (c) $O(1)$
- (d) $O(n \log n)$

****24. Which traversal gives sorted order in BST:****

- (a) Preorder
- (b) Postorder
- (c) Inorder
- (d) Level order

****25. Minimum height of binary tree with n nodes is:****

- (a) $\lfloor \log_2 n \rfloor$
- (b) $\lceil \log_2(n+1) \rceil - 1$
- (c) $n-1$
- (d) 1

**EXPRESSION EVALUATION & STACKS (Questions 26-35)**

****26. Minimum arithmetic operations to evaluate $f(x,y) = x^3 + 2x^2y + xy^2 + y^3$:****

- (a) 6
- (b) 8
- (c) 9
- (d) 10

****27. Postfix expression for $(A + B) * C - D / E$:****

- (a) $AB+C*DE/-$
- (b) $ABC+*DE/-$
- (c) $AB+C*D-E/$
- (d) $A+BC*DE/-$

****28. Prefix expression $--+abc*de$ equals:****

- (a) $((a+b-c)-d*e)$
- (b) $-(a+b-c)-d*e$
- (c) $-(-a+b*c)-d*e$
- (d) $-(-(a+b)*c)-(d*e)$

****29. For expression $A*B+C/D-E$, number of operands needed simultaneously:****

- (a) 2

- (b) 3
- (c) 4
- (d) 5

****30. Stack is used in:****

- (a) Function calls
- (b) Expression evaluation
- (c) Parsing
- (d) All of above

****31. Time complexity of postfix evaluation:****

- (a) $O(n)$
- (b) $O(n^2)$
- (c) $O(\log n)$
- (d) $O(n \log n)$

****32. Convert infix $(A+B)*(C+D)$ to prefix:****

- (a) $*+AB+CD$
- (b) $+AB*+CD$
- (c) $AB+CD+*$
- (d) $*AB+*CD$

****33. In postfix expression evaluation, when operator is encountered:****

- (a) Push to stack
- (b) Pop two operands and compute
- (c) Skip the operator
- (d) Convert to infix

****34. For infix to postfix conversion, operator priority matters for:****

- (a) Pushing to stack
- (b) Popping from stack
- (c) Both pushing and popping
- (d) Neither pushing nor popping

****35. Expression tree for $A*B+C$ has root:****

- (a) A
- (b) *
- (c) +
- (d) C

**SORTING & SEARCHING (Questions 36-45)**

****36. Time complexity of quicksort in worst case:****

- (a) $O(n \log n)$
- (b) $O(n^2)$
- (c) $O(n)$
- (d) $O(\log n)$

****37. Which sorting is stable:****

- (a) Quick sort
- (b) Heap sort
- (c) Merge sort
- (d) Selection sort

****38. Space complexity of merge sort:****

- (a) $O(1)$
- (b) $O(\log n)$
- (c) $O(n)$
- (d) $O(n \log n)$

****39. Heap sort worst case time complexity:****

- (a) $O(n \log n)$
- (b) $O(n^2)$
- (c) $O(n)$
- (d) $O(\log n)$

****40. For radix sort of 6 numbers, each 4-digit decimal, maximum comparisons:****

- (a) 240
- (b) 144
- (c) 180
- (d) 216

****41. Binary search requires:****

- (a) Sorted array
- (b) Unsorted array
- (c) Linked list
- (d) Stack

****42. Time complexity of linear search:****

- (a) $O(1)$
- (b) $O(\log n)$
- (c) $O(n)$
- (d) $O(n^2)$

****43. Which sorting has best average case performance:****

- (a) Bubble sort
- (b) Selection sort
- (c) Quick sort
- (d) Insertion sort

****44. Counting sort is efficient when:****

- (a) Range of input is small
- (b) Array is already sorted
- (c) Array size is small
- (d) Elements are unique

****45. In-place sorting algorithms:****

- (a) Merge sort
- (b) Quick sort
- (c) Counting sort
- (d) Radix sort

**COMPLEXITY ANALYSIS (Questions 46-55)**

****46. Arrange in increasing complexity order: $f_1(n)=2^n$, $f_2(n)=n^3$, $f_3(n)=n \log n$, $f_4(n)=n!$:**

- (a) f_3, f_2, f_1, f_4
- (b) f_1, f_2, f_3, f_4
- (c) f_4, f_1, f_2, f_3
- (d) f_3, f_2, f_4, f_1

****47. Time complexity of matrix multiplication ($n \times n$ matrices):**

- (a) $O(n^2)$
- (b) $O(n^3)$
- (c) $O(n)$
- (d) $O(n^4)$

****48. Space complexity of recursive factorial:**

- (a) $O(1)$
- (b) $O(\log n)$
- (c) $O(n)$
- (d) $O(n^2)$

****49. Which grows fastest:**

- (a) 2^n
- (b) $n!$
- (c) n^n
- (d) $n^{(\log n)}$

****50. Big-O of nested loop $i=1$ to n , $j=1$ to i :**

- (a) $O(n)$
- (b) $O(n \log n)$
- (c) $O(n^2)$
- (d) $O(n^3)$

****51. Master theorem $T(n) = 2T(n/2) + n$ gives:**

- (a) $O(n)$
- (b) $O(n \log n)$
- (c) $O(n^2)$
- (d) $O(\log n)$

****52. From string of length m , total substrings:**

- (a) m
- (b) m^2
- (c) $m(m+1)/2$

(d) 2^m

****53. Best case of bubble sort:****

- (a) $O(n)$
- (b) $O(n \log n)$
- (c) $O(n^2)$
- (d) $O(1)$

****54. Amortized analysis is used for:****

- (a) Average case complexity
- (b) Worst case over sequence of operations
- (c) Best case complexity
- (d) Space complexity

****55. Time complexity to multiply two $n \times p$ and $p \times m$ matrices:****

- (a) $O(npm)$
- (b) $O(n + p + m)$
- (c) $O(\max(n, p, m))$
- (d) $O(n^2pm)$

**ANSWER KEY - DATA STRUCTURES & ALGORITHMS**

****1-10:**** (a), (b), (b), (a), (c), (b), (c), (b), (b), (a)
****11-20:**** (a), (b), (c), (b), (b), (a), (b), (a), (a), (b)
****21-30:**** (a), (a), (b), (c), (b), (c), (a), (b), (a), (d)
****31-40:**** (a), (a), (b), (c), (c), (b), (c), (c), (a), (a)
****41-50:**** (a), (c), (c), (a), (b), (a), (b), (c), (b), (c)
****51-55:**** (b), (c), (a), (b), (a)

**Priority Study Areas for September 14th:**

1. ****Graph Algorithms (Q1-15):**** Practice MST, shortest path problems
2. ****Tree Operations (Q16-25):**** Focus on BST properties and traversals
3. ****Complexity Analysis (Q46-55):**** Master Big-O notation and growth rates
4. ****Expression Evaluation (Q26-35):**** Stack-based operations are frequently tested

****Time Allocation Suggestion:**** Spend 40% of your study time on this topic as it has highest weightage in ISRO pattern.